

RESEARCH INTEREST

- Adaptive design of clinical trials; Biostatistics; A/B testing; Spatial Experiment; Temporal Experiment.

EDUCATION

Sep. 2017– May. 2023	Department of Statistics, George Washington University Thesis: <i>Statistical Issues of Unobserved Covariates in Covariate-Adaptive Randomized Trial</i> . Ph.D. in Statistics. Advisor: Feifang Hu.	WASHINGTON D.C., U.S.
Sep. 2015 – Jun. 2017	Department of Statistics, George Washington University M.S. in Statistics	WASHINGTON D.C., U.S.
Sep. 2010 – Jul. 2014	School of Statistics, Renmin University of China B.S. in Statistics (including Risk Management and Actuarial Science)	BEIJING, CHINA

WORKING

Sep. 2023– present	Assistant Professor, Institute of Statistics and Big Data, Renmin University of China	BEIJING, CHINA
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PUBLICATIONS

1. Zhao, F., **Liu, Y.**, and Hu, F. (2025). Statistical Inference on the relative risk following covariate-adaptive randomization. *Biometrics* 81 (2) 1-12.
2. Guo, Z., **Liu, Y.**, and Xia, L. (2025). Covariate Selection for Optimizing Balance with Covariate-Adjusted Response-Adaptive Randomization. *Statistical Methods in Medical Research*. o(o) 1-29.
3. **Liu, Y.**, Xia, L., and Hu, F. (2025). Testing Heterogeneous Treatment Effect with Quantile Regression under Covariate-Adaptive Randomization. *Journal of Econometrics*. 249, 105808.
4. **Liu, Y.**, Zhou, Y., Li, P., and Hu, F. (2024). Cluster-Adaptive Network A/B Testing: From Randomization to Estimation. *Journal of Machine Learning Research*. 25 (170), 1-48.
5. **Liu, Y.**, and Hu, F. (2023). The Impact of Unobserved Covariates on Covariate-Adaptive Randomized Experiment. *The Annals of Statistics*. 51 (5) 1895 - 1920.
6. **Liu, Y.**, and Hu, F. (2022). Balancing Unobserved Covariates with Covariate-Adaptive Randomized Experiments. *Journal of the American Statistical Association*. 117 (538) 875-886.
7. **Liu, Y.**, Zhou, Y., Li, P. and Hu, F. (2022). Adaptive A/B test on Network with Cluster Structures. *In International Conference on Artificial Intelligence and Statistics*. PMLR. 151:10836-10851.
8. Oviedo, R. J., Nayak, T., **Liu, Y.**, Zhang, S., and Zhao, F. (2021). Laparoscopic Duodenal Switch Versus Roux-en-Y Gastric Bypass at a High-Volume Community Hospital: a Retrospective Cohort Study from a Rural Setting. *Obesity Surgery*, 31(2), 659-666.
9. **Liu, Y.**, and Gastwirth, J. L. (2020). On the capacity of the Gini index to represent income distributions. *METRON*, 78(1), 61-69.

SUBMITTED PAPERS

(* represent student)

1. **Liu, Y.**, Xia, L., and Hu, F. (2025+). Statistical Inference with Mixed-Effect Model for Covariate-Adaptive Randomized Experiment. *Journal of the American Statistical Association* (reject with resubmission).
2. Fu*, R. and **Liu, Y.** (2025+). Statistical Inference with Generalized Estimating Equations in Longitudinal Studies following Covariate-adaptive Randomization. *Statistica Sinica* (Major Revision).
3. Qin, Z., Hu, F., and **Liu, Y.** (2025+). To Adaptively Randomize or to Rerandomize: A Comparison of Covariate-Adaptive Randomization and Rerandomization. *Statistica Sinica* (Major Revision).

TEACHING EXPERIENCE

2022, 2023, 2024, 2025 Fall	Clinical Trial Methodology MS level, Instructor, Student Evaluation on average 99	BEIJING, CHINA
2023, 2024 Spring	Stochastic Process PhD level, Co-Instructor, Student Evaluation on average 96	BEIJING, CHINA
2023, 2024, 2025 Spring	Statistical Method in Biostatistics MS level, Co-Instructor, Student Evaluation on average 97.5	BEIJING, CHINA
2025 Spring	Data Collection PhD level, Co-Instructor, Student Evaluation: 99	BEIJING, CHINA
2025 Fall	Probability and Statistics Undergraduate level, Instructor	BEIJING, CHINA
Jun.2021 – Jul.2021	STAT 2112 Business & Econ Statistics II Undergraduate level, Instructor	WASHINGTON D.C., U.S.

SUPERVISED STUDENTS

Current PhD Students

Gufan, Jiang	(2025-)
Zhiyong, Liao	(2023-)
Ruichen, Fu	(2022-)

Current MS Students

Jiakun, Liu	(2025-)
Siyu, Liu	(2024-)
Ruimiao, Tang	(2024-)
Xueyan, Ding	(2024-)

Graduated MS Students

Xin, Guo	Position: Agricultural Bank of China (2023-2025)
Yuhao, Deng	Position: Zhejiang Provincial Bureau of Statistics (2023-2025)
Kun, Yi	Ph.D. in Statistics, HKUST (2023-2025)
Xingjian, Ma	Ph.D. in Statistics, U of Connecticut (2022-2024)
Jin, Mu	Ph.D. in Biostatistics, U of Wisconsin–Madison (2022-2024)
Zicheng, Xu	Position: Bank of Shanghai (2022-2024)

RESEARCH GRANT

- National Natural Science Foundation of China, Young Scientist Program: *Unobserved Covariates and Heterogeneity in Covariate-Adaptive Randomization*. Grant nos. 12301324. Period: 2024-2026, Principal Investigator.
- Meituan: *Design and Analysis of Credible A/B Tests with Food Delivery Platform — Balancing Regional Differences and Mitigating Spillover Effects* Period: 2023-2025, Principal Investigator.
- Meituan: *Temporal Adaptive Experiments — Balancing Temporal Features and Improving Estimation Accuracy*. Period: 2025-, Principal Investigator.
- Renmin University of China: *Research Grant for New Faculties* , Grant nos. 202330231 Period: 2023-2025, Principal Investigator.

AWARDS

Jun.2025	Research Collaboration – Outstanding Exploration Award	BEIJING, CHINA Meituan
Jun.2022	Outstanding Graduate Award	WASHINGTON D.C., U.S. Washington Statistical Society
Mar.2022	Fritz Scheurnen Award in Leadership and Service	WASHINGTON D.C., U.S. Department of Statistics, GWU
Aug.2020	Minna Kullback Award	WASHINGTON D.C., U.S. Department of Statistics, GWU
Aug.2019	Cornfield Endowment Fellowship,	WASHINGTON D.C., U.S. Department of Statistics, GWU
Jun.2019	GW Summer Pre-Dissertation Fellowship	WASHINGTON D.C., U.S. CCAS, GWU
Dec.2013	1 st Prize in Symphony Orchestra	BEIJING, CHINA The 4 th Beijing University Art Festival
Oct.2013	Distinguished Excellence Scholarship in Arts & Sports	BEIJING, CHINA Renmin University of China

CONSULTING

- Meituan: Working on improving the design of the A/B test for detecting spillover effect and the treatment effect.

SERVICE

2025– Present	Council Member, The Uniform Design Association of China, Chinese Mathematical Society	
2025– Present	Council Member, Division of Tourism and Big Data, Chinese Association for Applied Statistics	
2024– Present	Council Member, Association of Young Statisticians, Chinese Association for Industrial Statistics Teaching	
Dec. 2022– Present	Labor Union for ISBD at RUC, representative	BEIJING, CHINA

Dec. 2022– Present	Student Admissions Committee of ISBD at RUC, member	BEIJING, CHINA
Sep. 2021– Mar. 2022	PhD Student Committee, co-Chair Helping the department to organize student events, student seminars, helping the department to create curriculum for PhD students according to their need, etc.	WASHINGTON D.C., U.S.
Sep. 2017– Mar. 2022	PhD Student Committee, member Helping the department to organize student events, student seminars; helping the department to create curriculum for PhD students according to their need.	WASHINGTON D.C., U.S.
Sep. 2013– Jun. 2014	Chair of the Youth Orchestra of Renmin University of China Organize concerts, recordings; participate in competitions, etc.	BEIJING., CHINA

Journal Review

Biostatistics and Epidemiology
 Journal of Applied Statistics
 Journal of American Statistical Association
 Journal of Royal Statistical Society (series B)
 Statistics and Public Policy
 Statistica Sinica

PROFESSIONAL AFFILIATIONS

- Member, American Statistical Association
- Member, Institute of Mathematical Statistics
- Member, International Chinese Statistical Association
- Member, Chinese Mathematical Society

PRESENTATION

May. 2025	GWU Statistics Department 90 th Anniversary: “The Past, Present and Future of Statistics in the Era of AI”, “The Impact of Unobserved Covariates on Covariate-Adaptive Randomization.”
July. 2023	ICSA China Conference, “Balancing Unobserved Covariates with Covariate-Adaptive Randomization”.
March. 2022	AISTATS, “Adaptive A/B Test on Networks with Cluster Structures.”
Aug. 2020	JSM, “The Effect of Unobserved Covariate in Statistical Inference Under Covariate-Adaptive Randomized Experiment.”
Aug. 2019	JSM, “Unobserved Covariate Imbalance of Covariate-Adaptive Randomized Experiments.”